Intelligent Identification of Hazard-Based Red Zones, Carrying Capacity Assessment, and Immediate Relocation Needs for Vulnerable Habitations

PROBLEM STATEMENT ID SIH26191	**THEME** Disaster Management
CATEGORY Software	**TEAM NAME** Team PixelAlchemy
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Team Details

Member	Name	Role / Contribution
Team Leader	Dhrumi Vyas	Problem & Research Lead
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Problem & Unique Value Proposition

The Problem

- **Disaster-prone habitations are identified reactively rather than proactively**, with fragmented hazard data for floods, landslides, cloudbursts, and coastal erosion lacking a unified multi-hazard risk/red-zone assessment.
- **There is no systematic mechanism to prioritize vulnerable habitations for relocation** based on the severity and urgency of risk.
- **Relocation planning often overlooks the carrying capacity and suitability of safer destinations**, including water, healthcare, roads, schools, employment, land availability, and population pressure.

What Makes Us Different (USPs)

Existing tools are either visualization tools or policy guidelines, but do not connect hazard, population vulnerability, and site carrying capacity into an ordered actionable list. We represent the missing decision layer on top of the existing hazard-data ecosystem in India.

AI-Powered Multi-Hazard Red-Zone Detection

Predicts and combines multiple hazards using AI/ML and GIS to automatically identify Red, Orange, Yellow, and Safe zones.

Intelligent Proactive Relocation Planning

Identifies who is at risk, how many people need relocation, and the urgency through Immediate → Short-term → Medium-term → Monitor prioritization.

GIS + AHP + Carrying Capacity-Based Site Recommendation

Finds the safest relocation sites using GIS/AHP, assessing water, healthcare, education, roads, land availability, infrastructure, and population capacity to ensure the new location can support displaced people.

Proposed Solution

1) AI-Based Multi-Hazard Detection & Prediction

- Build an AI model that analyzes historical and real-time geospatial/environmental data to estimate the probability of different hazards.

2) Dynamic Multi-Hazard Risk Map

- Instead of separate hazard maps, combine the predictions into a single multi-hazard risk layer.
- The GIS map automatically displays : Red/Orange/Yellow/Green

3) Vulnerable Habitation Identification

- Identify actual villages/settlements/habitations that are inside dangerous zones.
- This makes the system useful to government/administrative decision-makers rather than only GIS analysts.

4) Population-at-Risk Estimation

- Once a red zone is identified, automatically calculate: Total population affected, No. of households, population density, vulnerable groups where data is available, No. of buildings, Critical facilities affected

5) AI-based Relocation priority

- Create a relocation priority index. Then classify settlements: Immediate, Short-term, Medium-term, Monitor

6) Automatic safe-site discovery

- After identifying a settlement requiring relocation, the system should automatically search for potential relocation areas.

7) GIS + AHP relocation suitability model & Relocation site carrying capacity assessment

- Use AHP to rank candidate locations against criteria based on safety, infrastructure, resources, and environmental factors, then verify each site's carrying capacity before recommending it. AHP Mathematical Formula → $(S = \sum_{i=1}^{n}(w_i \times c_i))$

Novelty and Uniqueness

EXISTING SOLUTIONS

- Bhuvan (ISRO) & GSI Maps: Provide static, single-hazard visualizations (like landslides) but **lack an actionable, decision-support engine for relocation.**
- IMD & Warning Systems: Offer reactive, time-bound hazard alerts but **do not evaluate long-term site carrying capacity for human habitation.**
- NDMA Guidelines: Provide comprehensive disaster policies but **remain static**, document-based frameworks rather than dynamic, data-driven software tools.

WHAT'S GENUINELY NEW

Unified Multi-Hazard Engine: Evaluates overlapping risks in one integrated platform, shifting from reactive mapping to *proactive risk assessment.*

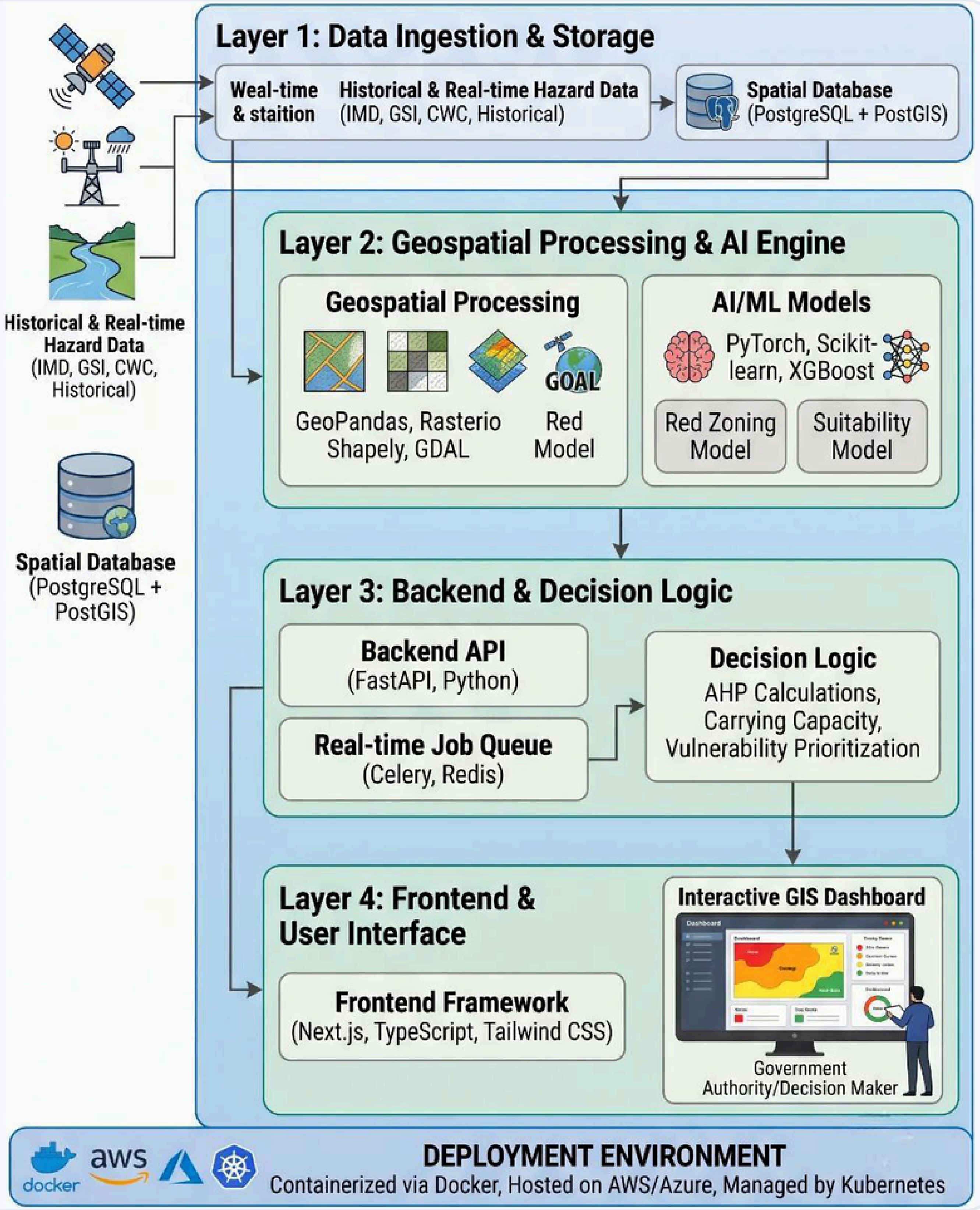
Holistic Carrying-Capacity Layer: Uniquely filters relocation sites based on safe terrain, infrastructure, and livelihood proximity - *not just "empty land."*

End-to-End Decision Support: Closes the gap by converting raw GIS data into a *transparent, auditable prioritization score* for authorities.

Technical Approach

TECH STACK

- Frontend : Next.js, Typescript
- UI : Tailwind CSS
- Interactive GIS : Interactive GIS
- Backend API : FastAPI, Python
- AI/ML : PyTourch, Scikit-learn, XGBoost
- Geospatial processing : GeoPandas, Rasterio, Shapely, GDAL
- Spatial database : PostgreSQL + PostGIS
- Real time backend jobs : Celery, Redis
- Deployment : Docker, AWS/Azure
- Hosting: Vercel, Render ; Kubernetes (Production)



Feasibility and Viability

Potential Challenges

- **Live Data Availability:** It is highly likely that real-time hazard streams and sensor feeds from federal authorities may be **restricted or behind paywalls** during hackathon prototyping.
- **Complex Land Rights and Resistance:** There may be complex land tenure, legal, and social issues if relocation is made compulsory.
- **False Alarms in Red Zoning:** A binary prediction model such as Red/Safe may lead to false alarms or failures, thus wasting resources.
- **Fragmented State Knowledge:** There is **fragmentation of data** standards, resolution, and hazard indicators across state disaster management authorities.

Mitigation Strategy

- **Modular Ingestion Pipeline:** Built upon a pluggable data ingestion architecture that has been **tested against simulated datasets** and is ready to be plugged into the IMD/CWC/GSI APIs.
- **Human-in-the-loop:** The system is a decision support system that calculates objective viability scores but **leaves final administrative decision-making to humans.**
- **Confidence Bands & Explainability:** Instead of relying on opaque binary judgments, the model **provides probabilistic confidence bands and breaks down the contributing factors** (e.g., rainfall, slope, exposure).
- **Region-Agnostic Schema:** A standardized and scalable schema that allows **easy ingestion of data from different sources** without redesigning the core architecture for each new state.

Impact and Benefits

Social Impact

- **Community Safety :** Protects vulnerable communities by identifying high-risk habitations early.
- **Fast Response :** Helps authorities prioritize relocation and emergency response.
- **Better Living Conditions:** Reduces loss of life, displacement, and repeated damage.

Economic Impact

- **Cost Saving :** Reduces disaster recovery and reconstruction costs through early, targeted relocation.
- **Market Opportunities :** Supports government agencies, NGOs & Disaster Management Organisations.
- **Revenue Potential:** Monetizable through SaaS subscriptions, Government licensing, and customized deployment.

Environmental Impact

- **Protects ecosystems:** Helps avoid settlements in environmentally sensitive and hazard-prone areas.
- **Promotes sustainable planning:** Supports climate-resilient land-use and relocation decisions.
- **Reduces resource loss:** Minimizes repeated reconstruction and associated material and energy consumption.

Research and References

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SECTION 08

Solution Summary

Criterion	Summary
Problem	Disaster-prone habitations are identified reactively, without unified multi-hazard red-zone mapping, risk-based relocation prioritization, or carrying-capacity assessment of safer destinations.
Proposed Solution	An AI-GIS platform that detects multi-hazard red zones, identifies and prioritizes vulnerable habitations, estimates population at risk, and recommends safe, capacity-verified relocation sites.
Novelty / USP	An end-to-end AI-GIS decision engine that unifies multi-hazard risk detection, population-based relocation prioritization, and carrying-capacity-verified site selection into one transparent, actionable system.
Feasibility	Buildable using existing satellite, GIS, weather, population and infrastructure datasets with proven AI/ML, AHP, and geospatial technologies.
Impact	Protects vulnerable communities, reduces disaster recovery costs, and enables authorities to make faster, data-driven relocation decisions.

Mockup UI & Illustrative Representation:

Link - PIXELALCHEMY_SIH191